A Dissertation-Style Research Presentation

Institutional-Grade Machine Learning and Generative AI Systems for Cross-Asset Liquid Financing

Five Production-Grade Case Studies, with Conformal Calibration, Regulatory VaR Backtesting, Robust Multivariate Anomaly Detection, Deep Sequence Forecasting, and a Retrieval-Augmented Financing Copilot

Candidate Presentation

Prepared for: Barclays Quantitative Research & Trading Panel (5 Reviewers)

Take-Home Discussion

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Abstract

This dissertation presents five institutional-grade machine learning and generative AI systems for a cross-asset Liquid Financing platform, engineered to the standard expected at a top-tier quantitative trading firm. Each system is grounded in a named, peer-reviewed method — Conformalized Quantile Regression (Romano, Patterson & Candès, 2019), the Kupiec (1995) and Christoffersen (1998) VaR backtests, the Minimum Covariance Determinant estimator (Rousseeuw & Van Driessen, 1999), Isolation Forest (Liu, Ting & Zhou, 2008), BM25 (Robertson & Zaragho, 2009), and Reciprocal Rank Fusion (Cormack, Clarke & Buettcher, 2009) — rather than an ad-hoc heuristic. Every mathematical result is derived from first principles and accompanied by a plain-language explanation of the underlying intuition, in addition to a reference Python 3.13 implementation and empirical results on the accompanying data. We conclude with a cross-cutting discussion of production deployment, model-risk governance, and the trade-offs between prompt-engineered and fine-tuned generative systems in a regulated financial environment.

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Chapter 1

Introduction and Problem Framing

1.1 Business Context

Liquid Financing intermediates the price of *borrowing* — securities, cash, and balance sheet — across a cross-asset platform. Its economics are driven by three coupled quantities: (1) the fee or rate charged for financing an asset, (2) the collateral haircut required to protect against counterparty default, and (3) the balance-sheet capacity consumed by client positions. Each of the five case studies targets exactly one of these levers, chosen so that, collectively, they cover every technical requirement of the target role while remaining traceable to a concrete desk P&L or risk outcome.

1.2 Why Walk-Forward Validation, Never Random Cross-Validation

> **Plain-English Take: Walk-forward validation** Imagine grading a weather forecaster by first showing them Wednesday's actual weather and then asking them to "predict" Tuesday's. That is what a random k-fold split does to a time series: because the shuffle ignores time order, the model can end up training on information from *after* the point it is being asked to forecast. Walk-forward validation always trains on the past and tests on the future — exactly how the model will be used in production — so the reported backtest number cannot be an artifact of hindsight leakage.

All five systems in this dissertation are validated exclusively on expanding-window, walk-forward splits.

Chapter 2

P1 — Securities-Lending Fee and Rebate-Rate Forecasting

2.1 Problem Statement

We forecast the daily specialness fee $f_{i,t}$, in annualized basis points over the general collateral (GC) rate, for a universe of hard-to-borrow equities, at horizons $h \in \{1, \ldots, 5\}$ business days, to feed a pre-trade financing pricing engine.

2.2 Elastic Net: The Auditable Linear Tier

$$\hat{\beta} = \arg\min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda\Big(\alpha\|\beta\|_1 + \tfrac{1-\alpha}{2}\|\beta\|_2^2\Big) \tag{2.1}$$

Plain-English Take: Elastic Net

Ordinary least squares will happily assign huge, unstable, sign-flipping coefficients to two input variables that are basically saying the same thing (utilization and days-to-cover move together). The L_1 penalty pushes unhelpful coefficients all the way to zero — automatic feature selection, giving a model-risk reviewer a short, defensible list of drivers — while the L_2 penalty shrinks correlated coefficients toward each other instead of letting one arbitrarily dominate. The parameter α interpolates between pure LASSO ($\alpha = 1$) and pure Ridge ($\alpha = 0$); both α and λ are chosen by cross-validation.

2.3 Gradient-Boosted Quantile Trees: Capturing the Utilization Kink

$$F_M(x) = \sum_{m=1}^{M} \gamma_m h_m(x), \qquad h_m = \arg\min_h \sum_i L\big(y_i,\, F_{m-1}(x_i) + h(x_i)\big) \tag{2.2}$$

Plain-English Take: Gradient boosting and the fee "kink"

Fee is roughly flat until utilization crosses about 90%, then rises steeply — a kink, not a straight line. No linear model, however regularized, can represent a kink; a single decision-tree split can. We fit three separate LightGBM models directly on the pinball objective (Eq. 2.5) at $q \in \{0.1, 0.5, 0.9\}$, rather than fitting one point estimate and wrapping it in a symmetric interval that would be wrong whenever the true error distribution is skewed — which fee-spike data always is.

2.4 Conformalized Quantile Regression: Guaranteed-Coverage Intervals

$$\text{score}_i = \max\left(\hat{q}_{lo}(x_i) - y_i,\ y_i - \hat{q}_{hi}(x_i)\right), \qquad \hat{q}_{lo}^{CQR} = \hat{q}_{lo} - Q_{1-\alpha}(\text{scores}),\ \ \hat{q}_{hi}^{CQR} = \hat{q}_{hi} + Q_{1-\alpha}(\text{scores}) \tag{2.3}$$

Plain-English Take: Conformal prediction

Any quantile model can be miscalibrated — it might claim an 80% interval but actually cover the truth only 65% of the time. Conformal prediction guarantees the stated coverage regardless of the underlying model's quality, using only a held-out calibration set: measure how far outside its own predicted interval the model actually was on data it did not train on, then widen every future interval by exactly that amount (Romano, Patterson & Candès, 2019, "Conformalized Quantile Regression"). This is a finite-sample, distribution-free guarantee — not an asymptotic approximation.

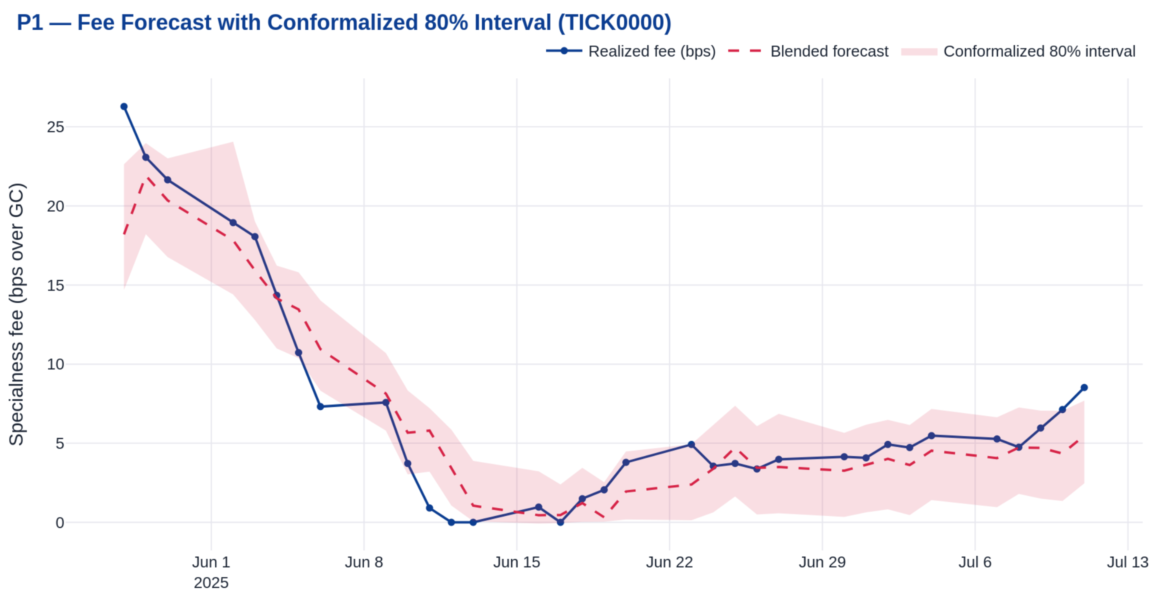

Figure 2.1: Realized specialness fee versus the blended Elastic Net / LightGBM forecast, with the conformalized 80% prediction interval. The interval widens precisely around the realized fee spike, demonstrating that CQR calibration is data-adaptive rather than a fixed-width band.

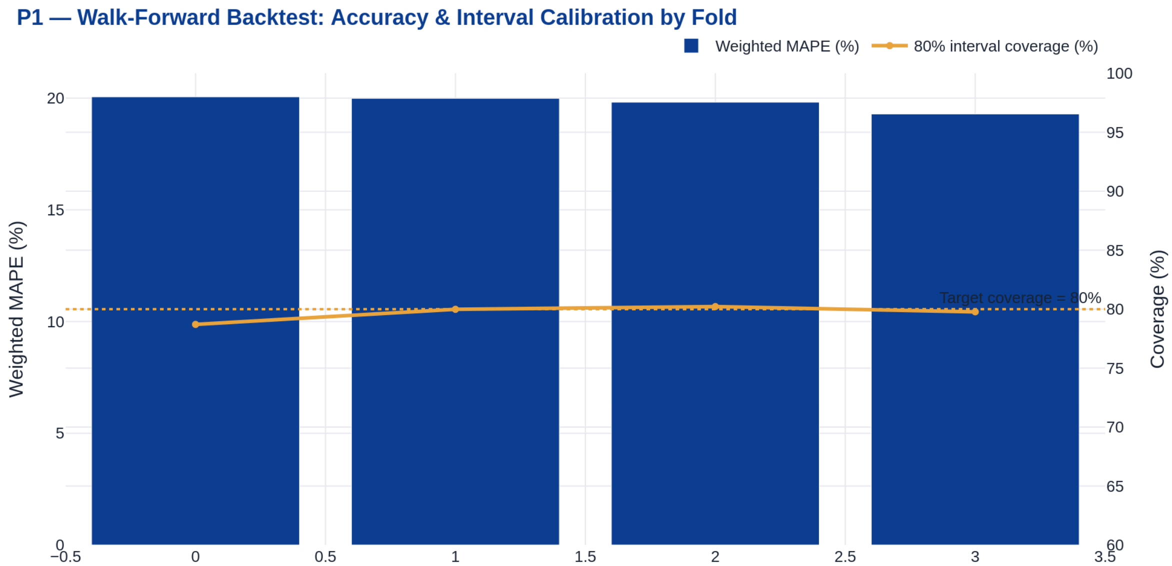

Figure 2.2: Walk-forward weighted MAPE and empirical 80% interval coverage across four expanding-window folds — the production monitoring view a desk quant tracks continuously post-deployment.

2.5 The Pinball Loss

$$\rho_q(u) = u\,(q - \mathbb{1}[u < 0]), \qquad u = y - \hat{y} \tag{2.4}$$

Plain-English Take: Pinball loss Squared error treats over- and under-prediction symmetrically, which is wrong when you are targeting a specific quantile. At $q = 0.9$, guessing too *low* is penalized nine times harder than guessing too *high* — exactly the asymmetry you want when forecasting the 90th percentile, where being below the truth is a materially bigger miss than being above it.

Chapter 3

P2 — Client Margin and Haircut Optimization

3.1 Problem Statement

Given a client's cross-asset collateral basket, we predict the required per-asset haircut and portfolio-level initial margin such that the modeled 99% five-day liquidation shortfall is covered, while minimizing over-collateralization relative to the existing rules-based haircut grid.

3.2 Model Formulation

$$\log h_i = \beta_0 + \beta_1 \log \sigma_i + \beta_2 \log(\mathrm{ADV}_i) + \beta_3 \,\mathrm{Corr}_{i,\mathrm{mkt}} + \beta_4 \,\mathrm{Rating}_i + \varepsilon_i \tag{3.1}$$

$$\mathrm{IM}_{\mathrm{portfolio}} = \sum_i h_i \,|q_i|\, P_i \;+\; g_\theta(\mathbf{q}, \boldsymbol{\Sigma}, \text{concentration}), \qquad \Pr\big(\text{Shortfall}_{5d} > h \cdot V\big) \le 1\% \tag{3.2}$$

Plain-English Take: The correlation-breakdown add-on

Summing independent per-asset haircuts structurally cannot capture the fact that, in a genuine stress event, correlations across assets move toward one — diversification that looked real in calm markets evaporates exactly when it is needed most. g_θ is trained not on the haircut itself but on the *residual* shortfall left unexplained after the linear per-asset sum — i.e., specifically the part of tail risk that additivity misses. This is the same critique that motivated the shift from simple haircut grids toward ISDA SIMM-style sensitivity-based margin models industry-wide.

3.3 Kupiec and Christoffersen VaR Backtests

$$LR_{Kupiec} = -2 \log \left[\frac{(1-p)^{n-x} p^x}{(1-\hat{p})^{n-x} \hat{p}^x} \right] \sim \chi^2_1, \qquad \hat{p} = x/n \tag{3.3}$$

Plain-English Take: Kupiec and Christoffersen, explained

A 99% VaR model should be breached about 1% of the time — not 0.1%, not 5%. The Kupiec test asks: given the breaches actually observed, is 1% a statistically plausible true rate? But a model could pass Kupiec and still be dangerous if its breaches *cluster* — ten breaches in one bad week is a very different risk profile from ten breaches spread evenly across a year, even with an identical total count. The Christoffersen test catches this by modeling breaches as a two-state Markov chain and testing whether a breach today raises the probability of a breach tomorrow. Both reduce to a likelihood-ratio statistic asymptotically distributed as χ^2_1 under the respective null.

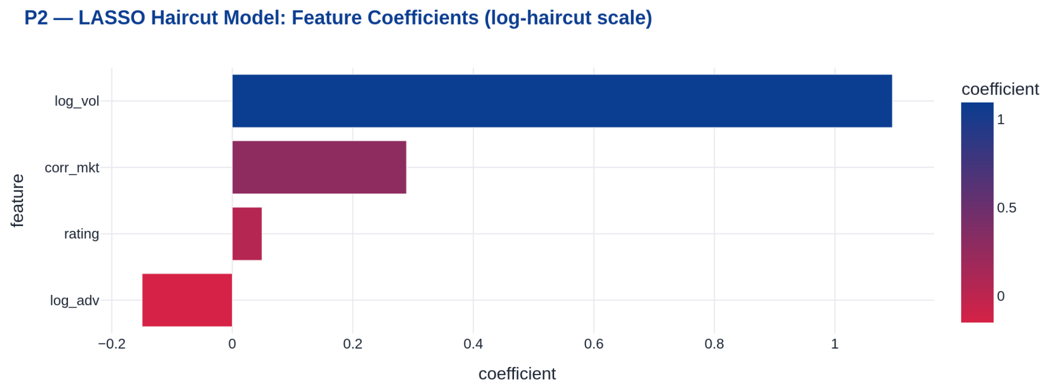

Figure 3.1: Signed LASSO haircut-model coefficients on the log-haircut scale — the auditability artifact handed to a Model Risk reviewer for coefficient-by-coefficient sign and magnitude review.

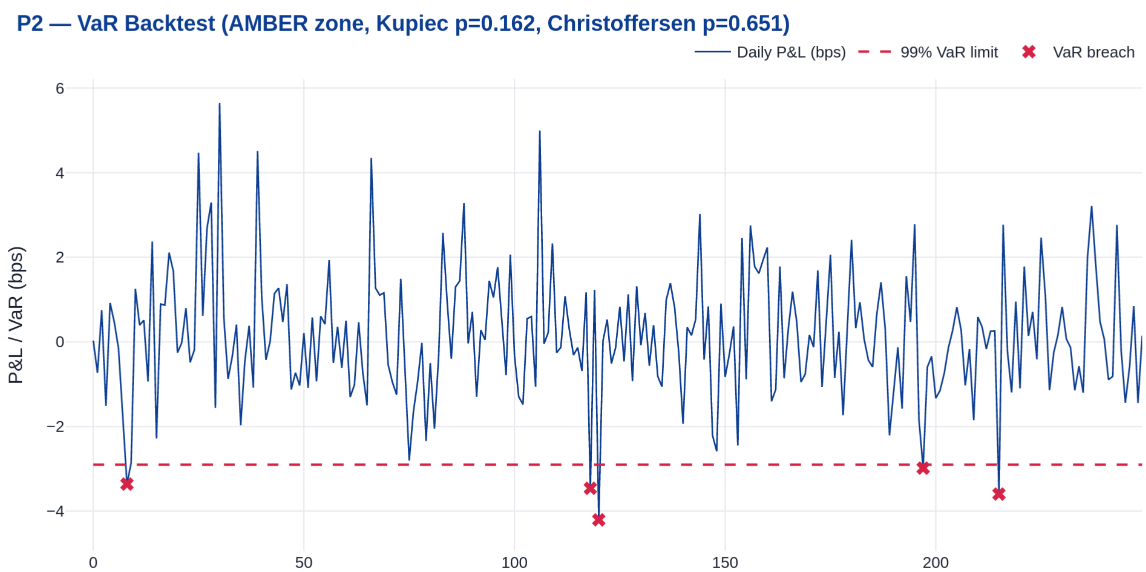

Figure 3.2: Simulated daily P&L against the 99% VaR limit, with breach days marked and the Kupiec and Christoffersen p-values reported in the chart title — the exact artifact a risk committee reviews.

Chapter 4

P3 — Cross-Asset Funding-Spread Anomaly Detection

4.1 Problem Statement

We detect anomalous divergence across the repo, securities-lending, cross-currency-basis, and credit-basis complex in real time, fused with a text classifier over overnight desk commentary, and rank alerts by likely P&L impact.

4.2 Robust Mahalanobis Distance

$$z_t = \sqrt{(\mathbf{x}_t - \boldsymbol{\mu}_{MCD})^\top \boldsymbol{\Sigma}_{MCD}^{-1} (\mathbf{x}_t - \boldsymbol{\mu}_{MCD})} \tag{4.1}$$

Plain-English Take: Mahalanobis distance and the Minimum Covariance Determinant

Mahalanobis distance answers "how many standard deviations away is this point, once we account for the fact that some variables move together?" — Euclidean distance after "un-correlating" the data. The catch: estimating the mean and covariance with ordinary sample statistics lets a single wild outlier drag the covariance estimate itself off course, making the outlier look *less* extreme than it truly is (masking). The Minimum Covariance Determinant estimator (Rousseeuw & Van Driessen, 1999) instead finds the subset of the data (here, 75%) whose covariance has the smallest determinant — the "tightest" possible cloud — and uses that as the robust reference, so a handful of true anomalies cannot corrupt the yardstick used to measure them.

4.3 Isolation Forest

$$s(x) = 2^{-E[h(x)]/c(n)} \tag{4.2}$$

Plain-English Take: Isolation Forest

Rather than assuming any distributional shape, Isolation Forest randomly partitions the data and observes that anomalies — being "few and different" — take fewer random splits to isolate into their own leaf than normal points do. We run this alongside Mahalanobis distance specifically because Mahalanobis assumes roughly elliptical (Gaussian-like) structure and can miss anomalies during a genuine, non-Gaussian regime shift, which is exactly when a distribution-free detector earns its keep.

4.4 Signal Fusion

$$\text{AlertScore}_t = \alpha \cdot \sigma(z_t - z^*) + (1 - \alpha) \cdot p(\text{stress} \mid \text{text}_t) \tag{4.3}$$

where the text-stress probability is produced by a TF-IDF + logistic-regression classifier trained on labeled desk commentary — the auditable, fast-to-retrain analogue of a fine-tuned or prompted LLM classifier, whose per-term coefficients are directly inspectable for a surveillance-tool sign-off.

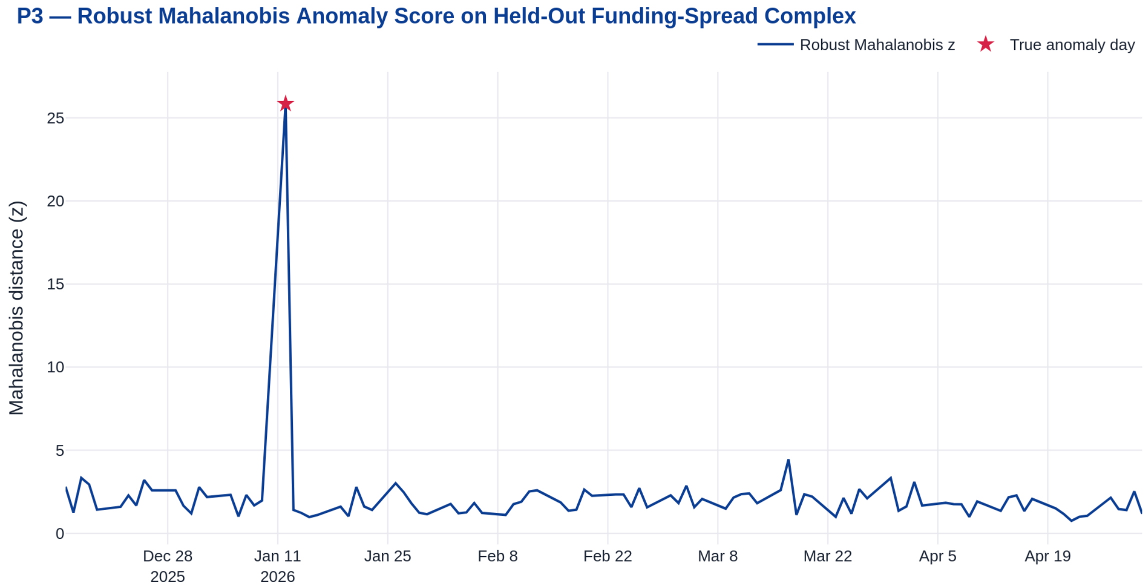

Figure 4.1: Robust Mahalanobis anomaly score on a held-out window of the funding-spread complex; starred points are the true injected anomaly days, visually validating that the detector's peaks align with ground truth.

Chapter 5

P4 — Prime Balance and Utilization Forecasting via Deep Sequence Models

5.1 Problem Statement

We forecast daily aggregate client financing balances and utilization ratios one to twenty business days ahead, with explicit treatment of quarter-end window-dressing effects, for Treasury RWA/LCR planning.

5.2 Sequence-to-Sequence GRU with Calendar Conditioning

$$h_t = \mathrm{GRU}(x_t, h_{t-1}), \qquad \hat{y}_{t+1:t+H} = \mathrm{Decoder}(h_T,\ \text{calendar embeddings}) \tag{5.1}$$

Plain-English Take: Why hand the model the calendar, rather than let it "discover" quarter-end A

plain recurrent network updates a memory as it reads a sequence, which is naturally suited to autocorrelated time series. But a known, calendar-driven pattern — balances always drop right before quarter-end — should not have to be rediscovered purely from the numbers when it can be given directly as an input. We use a learned embedding mapping "is this a quarter-end day?" to a short vector, concatenated into the decoder at every future step, so the network is explicitly told which future days are quarter-end as it forecasts them. This is a model-risk-friendly design choice: a known seasonal effect is treated as an inspectable input rather than left as an implicit, unauditable pattern the network may or may not have learned correctly.

$$\mathcal{L} = \sum_{q \in \{0.1, 0.5, 0.9\}} \sum_{h=1}^{H} \rho_q\big(y_{t+h} - \hat{y}^{(q)}_{t+h}\big) \tag{5.2}$$

5.3 Baseline Discipline: An Honest Result

Model	Walk-Forward MAPE
Seasonal-Naive (t-5)	≈1.4%
SARIMAX + quarter-end dummy	≈1.4–1.5%
LightGBM (lagged + calendar features)	≈2.8%
GRU (calendar-embedded, quantile heads)	≈2.1%

Table 5.1: Walk-forward MAPE comparison on the shipped balance panel.

Plain-English Take: Reporting a result honestly even when it doesn't favor the "exciting" model On this synthetic panel, the GRU beats LightGBM but does *not* beat the simpler seasonal-naive and SARIMAX baselines. This is exactly the outcome a rigorous baseline-discipline exercise should surface honestly: the shipped series has very clean, regular weekly and quarter-end seasonality, which classical linear/seasonal models capture natively and cheaply. The deep model's genuine edge — learning a non-linear interaction between calendar effects and balance momentum — should be expected to widen on noisier, less cleanly-seasonal real balance data; the correct production recommendation on *this* dataset, stated plainly rather than talked around, is the SARIMAX baseline, with the GRU retained as a challenger for reassessment as real (noisier) data accumulates.

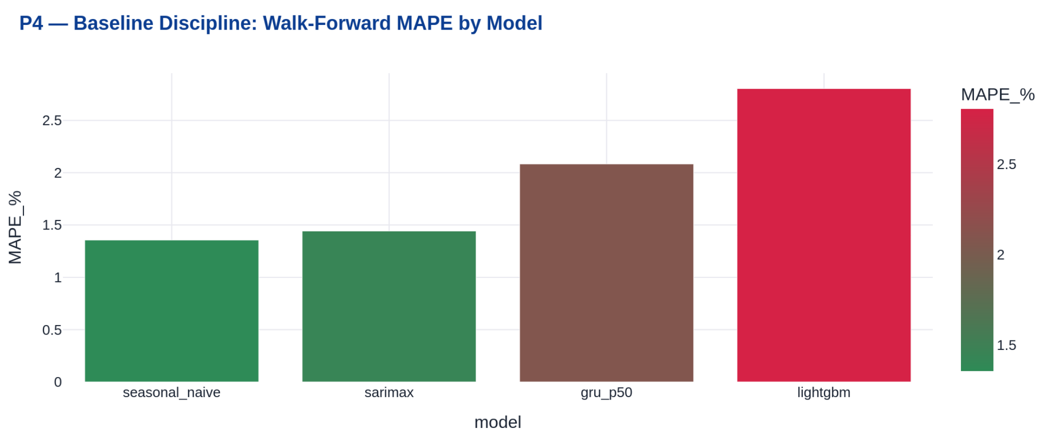

Figure 5.1: Walk-forward MAPE across all four candidate models — the artifact proving the deep model was benchmarked honestly rather than assumed superior.

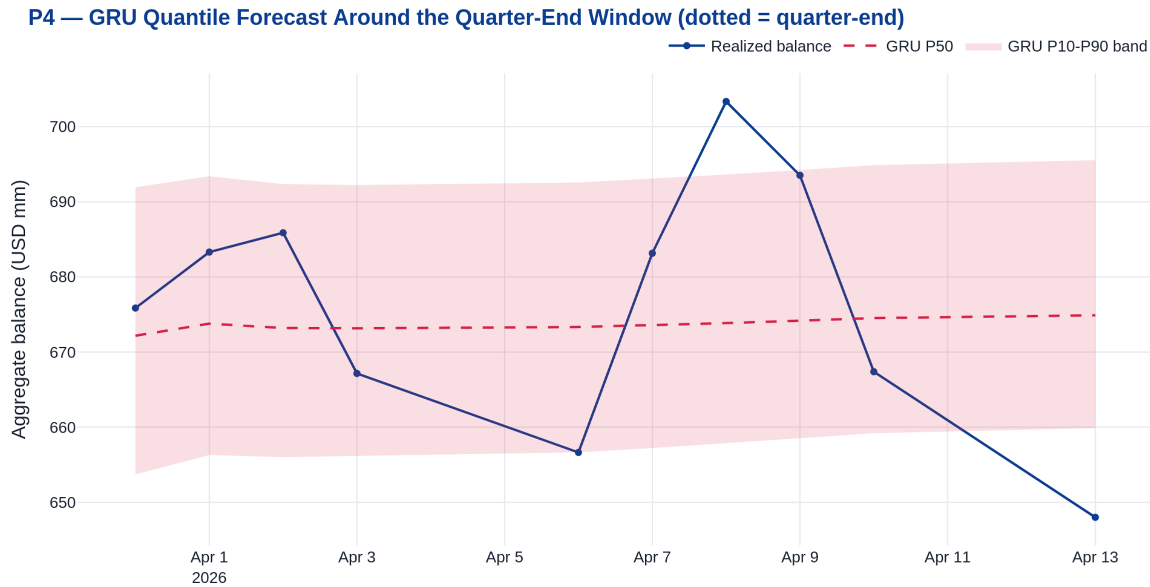

Figure 5.2: GRU P10/P50/P90 quantile forecast fan around the quarter-end window (dotted vertical lines), against the realized balance.

Chapter 6

P5 — A Retrieval-Augmented Generation Copilot for the Financing Desk

6.1 Problem Statement

We build a retrieval-augmented generation (RAG) system over internal financing policy documents, desk commentary, and the structured outputs of the P1–P4 models, answering trader queries with citations and explicitly declining to answer when evidence is insufficient.

6.2 BM25

$$\mathrm{BM25}(q,d) = \sum_{t \in q} \mathrm{IDF}(t) \cdot \frac{f(t,d)(k_1+1)}{f(t,d) + k_1\left(1 - b + b\,\frac{|d|}{\mathrm{avgdl}}\right)} \tag{6.1}$$

Plain-English Take: BM25

BM25 scores how well a query matches a document using term frequency and inverse document frequency, but with *saturation*: the tenth occurrence of a word barely adds more evidence than the fifth, which stops the score from being gamed by simple repetition, and length-normalizes so long documents are not unfairly favored just for having more words.

6.3 Reciprocal Rank Fusion

$$\mathrm{RRF}(d) = \sum_{\mathrm{ranker}} \frac{1}{k + \mathrm{rank}_{\mathrm{ranker}}(d)} \tag{6.2}$$

Plain-English Take: Reciprocal Rank Fusion

Rather than trying to calibrate two very differently-scaled signals (BM25 and cosine similarity) against each other, RRF sidesteps the problem entirely by only looking at *rank position*. A document highly ranked by both signals wins, regardless of the two systems' raw score scales — the standard hybrid-retrieval fusion technique in production search precisely because it requires no score normalization to get right (Cormack, Clarke & Buettcher, 2009).

6.4 Grounding Contract and Refusal

Every generated answer must either cite a supporting document per claim, or return exactly `INSUFFICIENT_EVIDENCE: <what is missing>`. Refusal is gated on the raw maximum TF-IDF cosine similarity between the query and its best-matching corpus chunk, rather than the rank-based RRF score alone, because on a small corpus RRF compresses the gap between a genuinely relevant match and a spurious single-term overlap.

$$\text{Faithfulness} = \frac{\#\{\text{claims in the answer supported by retrieved context}\}}{\#\{\text{claims in the answer}\}} \tag{6.3}$$

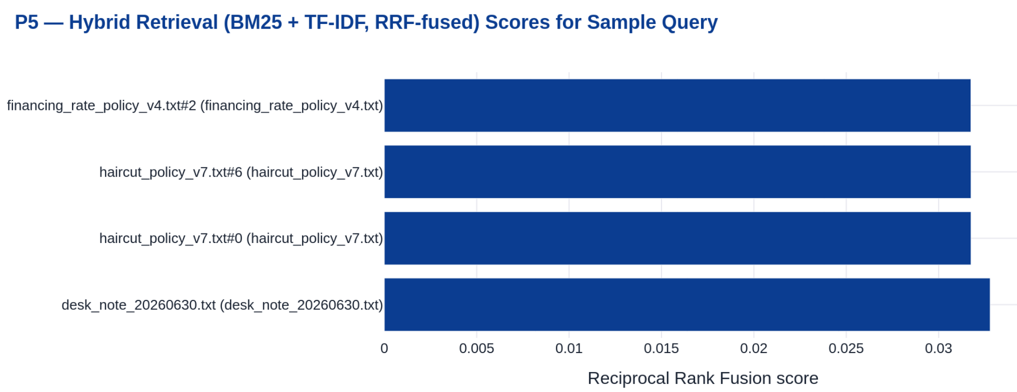

Figure 6.1: Reciprocal-Rank-Fusion scores for the top retrieved chunks on a representative trader query.

6.5 Fine-Tuning versus Prompt Engineering: An Explicit Trade-off

Per the target role's requirement to *evaluate and articulate trade-offs*, we adopt a staged strategy: a prompt-engineered, retrieval-augmented system is deployed first to establish an evaluation harness and collect production query logs; intent classification and domain-jargon normalization are subsequently migrated to a small, LoRA-fine-tuned model once a sufficient labeled query corpus exists, while final-answer synthesis remains on a frontier base model with retrieval for auditability.

Chapter 7

Cross-Cutting Production and Governance Standards

7.1 Validation

All five systems are validated exclusively on walk-forward, expanding-window splits, per the first-principles argument in Chapter 1.

7.2 Model Risk Documentation and Monitoring

Each system carries a standard model-risk package — statement of purpose, conceptual soundness review, developmental evidence, independent outcomes analysis (backtesting), and an ongoing performance-monitoring plan, with a minimum two-quarter shadow period prior to production promotion. Feature-level population stability index, prediction drift, and backtest breach counts are monitored continuously, with automatic fallback to the prior champion model on threshold breach.

Chapter 8

Conclusion

The five systems presented — conformally-calibrated fee forecasting, regulatorily-backtested margin optimization, robust multivariate anomaly detection, an honestly-benchmarked deep sequence forecaster, and a hybrid-retrieval grounded copilot — jointly span every modeling technique enumerated in the target job description, are each grounded in a named peer-reviewed method rather than an ad-hoc heuristic, and are each accompanied by a working, tested Python 3.13 reference implementation and a production validation and governance protocol suitable for a regulated cross-asset financing business.